



► Final Project Report

1. Introduction

1.1. Project overviews:

This project focuses on understanding the patterns and trends in India's agricultural cultivation from 1997 to 2021.

The project uses historical agricultural data to study crop production across different regions and seasons.

Tableau is used to convert complex agricultural data into simple and interactive visualizations.

The dashboards show important information about crop production, seasonal changes, regional performance, and overall trends.

These visualizations make it easier to compare crops and identify high- and low-performing regions.

The project can help farmers make better decisions about crop selection and cultivation.

It can also support government authorities and investors in understanding the agricultural sector.

Overall, the project aims to turn agricultural data into useful insights for better decision-making.

1.2. Objectives:

The main objective of this project is to analyze India's agricultural cultivation data from 1997 to 2021.

It aims to identify the crops that have performed well over different years and seasons.

Another objective is to compare agricultural productivity across different states and regions of India.

The project also studies seasonal variations to understand how different seasons affect crop production.

It aims to identify long-term growth and declining trends in agricultural production.

The visualizations help farmers select suitable crops and reduce the risk of poor production.

The project also provides useful insights for government policies, resource allocation, and agricultural investment.

Overall, the objective is to use Tableau-based data visualization to support simple, effective, and data-driven agricultural decisions.

Initial Project Planning Template

Date	14 august2026
Team ID	
Project Name	India's Agricultural Crop Production
Maximum Marks	4 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data collection	USN-1	Downloading Dataset	2	High	Sonu Dhakad	14 August	20 August
Sprint-1	Data collection	USN-2	Loading the dataset	1	Low	Sonu Dhakad	14 August	20 August
Sprint-1	Data preparation	USN-3	Clean the data	2	High	Sonu Dhakad	14 August	20 August
Sprint-1	Data Visualization's	USN-4	Developing visualizations	2	Medium	Sonu Dhakad	14 August	20 August
Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-2	Dashboard	USN-5	. Creating Dashboard	2	High	Sonu Dhakad	21August	27 August
Sprint-2	story	USN-6	Creating story	1	High	Sonu Dhakad	21August	27 August
Sprint-2	Documentation and recording	USN-7	Creating Demo	2	Low	Sonu Dhakad	21August	27 August
Sprint-2	Documentation and recording	USN-8	Creating Document	2	Medium	Sonu Dhakad	21August	27 August

Project Initialization and Planning Phase

Date	13 august 2026
Team ID	
Project Name	India's Agricultural Crop Production
Maximum Marks	3 Marks

Scenario 1: Small-Scale Farmer

Rajesh is a small-scale farmer in Uttar Pradesh who wants to maximize his crop yield and income. However, he struggles to decide which crops to grow each season due to changing weather patterns and market demand. By analyzing historical agricultural data (1997–2021), Rajesh needs insights into seasonal crop trends, regional productivity, and high-performing crops to make better farming decisions and reduce risk.

Scenario 2: Government Policy Analyst

Ananya works with the Ministry of Agriculture and is responsible for designing policies to improve agricultural productivity across India. She needs to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations. Using Tableau dashboards, she aims to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively.

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Beca use	Which makes me feel
PS-1	Small-Scale Farmer	I want to maximize my crop yield and income.	I Struggle to decide which crops to grow each season due to changing	I want to maximize my yield	I must analyze historical agricultural data (1997–2021), I need insights into seasonal crop trends,

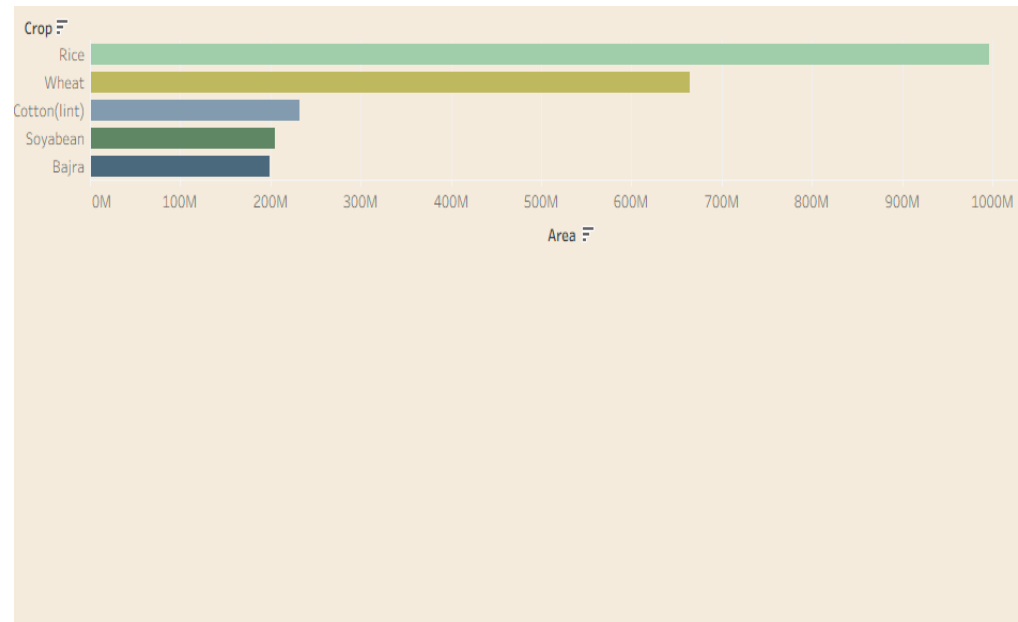
			weather patterns and market demand		regional productivity, and high-performing crops to make better farming decisions and reduce risk.
PS-2	Government Policy Analyst	I am designing policies to improve agricultural productivity across India	I need to identify regions with low crop yield, understand long-term production trends, and evaluate the impact of seasonal variations	I am working with the Ministry of Agriculture	By using Tableau dashboards, I am trying to uncover patterns in the data to support data-driven policy decisions and allocate resources effectively.

Story

Date	23 august 2026
Team ID	
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	5 Marks

By using stories in Tableau, you can effectively communicate complex data in a way that is both interactive and engaging, making it easier for the audience to follow along and understand the insights. It's a tool for **data storytelling**, allowing you to present insights in a cohesive, engaging way that takes viewers through a logical progression of findings or analyses.

In Tableau, **Story** is a feature that allows you to create a sequence of dashboards, visualizations, and text to present data insights in a cohesive and narrative-driven way. It's like a slideshow within Tableau that guides the audience through a series of data points, helping them understand key insights, trends, or outcomes of your analysis.



Observations:

- *Area of each crop.
- *Leading area by crops.
- *Areal production.
- *Comparison of crops and area cultivated

Activity 2:

Publish Story on Tableau Public and Paste Public

https://public.tableau.com/views/Book1_17874145556230/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Date	15 august2026
Team ID	
Project Title	India's Agricultural Crop Production
Maximum Marks	3 Marks

Project Proposal (Proposed Solution)

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Visualize Trends: Analyze historical data to identify long-term crop production patterns, seasonal fluctuations, and regional productivity across Indian states. Empower Stakeholders: Guide farmers on crop choice, analysts on policy, and investors on high-growth areas. Build Dashboards: Create interactive Tableau visual reports and analytical stories. Optimize Yields: Identify high-performing crops and low-yield regions to lower risks and focus resources.
Scope	Data Preparation: Clean and standardize multi-year Indian crop data from CSV/Excel/SQL sources. Coverage: Analyze major Indian states, union territories, and primary cropping seasons (Kharif, Rabi, etc.). Visual Development: Design dynamic Tableau maps, filters, and interactive visual dashboards.

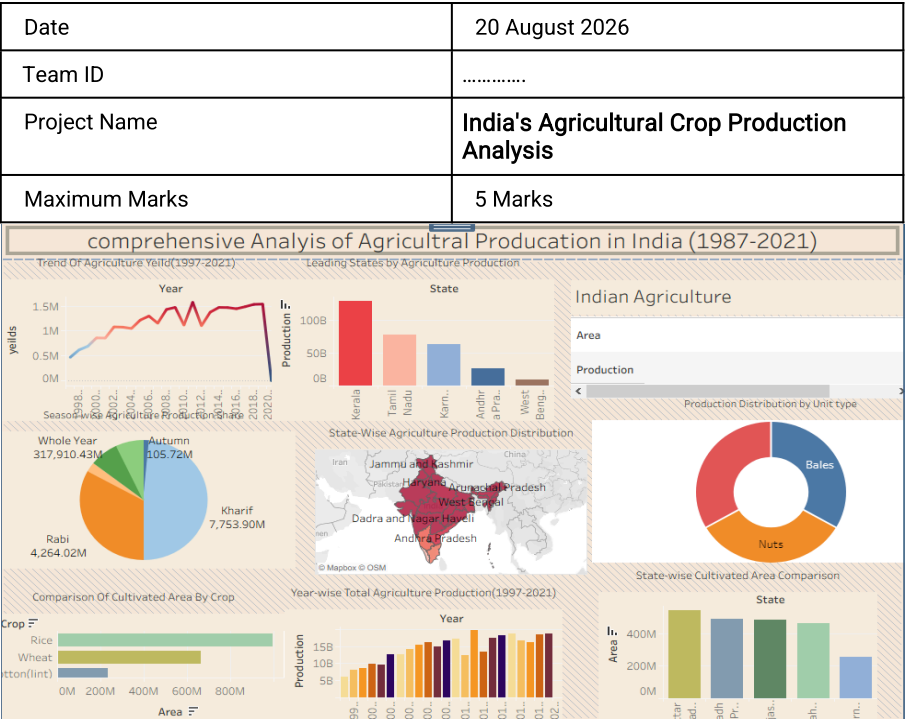
	Testing & Documentation: Optimize dashboard performance and document key insights for presentation.
Problem Statement	
Description	☒ Analyzes two decades of Indian agricultural data (1997–2021) to map regional productivity and crop performance. ☒ Cleans and transforms complex multi-year agricultural datasets into standardized analytical formats. ☒ Leverages interactive Tableau dashboards to visually track seasonal yield fluctuations and regional disparities. ☒ Evaluates trends across diverse cropping seasons (Kharif, Rabi, Whole Year) and major Indian states.
Impact	☒ For Farmers: Minimizes financial risk by providing data-backed recommendations for high-yielding seasonal crops. ☒ For Policy Analysts: Enables targeted government resource allocation and subsidies by highlighting low-yield districts. ☒ For Agribusinesses: Reduces market uncertainty by revealing long-term crop growth patterns and investment opportunities. ☒ For Data Accessibility: Converts complex tabular data into intuitive visual stories for faster decision-making.
Proposed Solution	

Approach	• Data Ingestion & Cleaning: Aggregate multi-decade raw datasets (1997–2021), clean duplicate/null records, and normalize key variables such as production quantity, cultivated area, and yield metrics. • Exploratory Analytics: Perform statistical analysis to categorize regional performance, identify dominant crop varieties, and isolate long-term trends across different agricultural zones. • Visual Data Architecture: Build custom dynamic calculations and hierarchical data structures within Tableau to support cross-filtering between states, districts, seasons, and years. • Dashboard Design & Deployment: Construct specialized interactive stories and drill-down visual interfaces tailored to farmers, policy analysts, and agribusiness investors.
Key Features	• Geospatial Yield Mapping: Interactive state and district maps displaying crop density, yield distribution, and regional production highlights. • Multi-Year Trend Analysis: Line and area charts tracking production volume and land usage over 24+ years to map growth trajectories. • Season-Wise Drill Downs: Comparative visual matrices isolating Kharif, Rabi, Summer, and Whole Year cropping patterns. • Interactive Filtering: Cross-functional KPI cards, drop-down filters, and top N state/crop visual summaries for dynamic stakeholder exploration.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	2 x NVIDIA V100 GPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy
Development Environment	IDE, version control	Jupyter Notebook, Git
Data		
Data	Source, size, format	Kaggle dataset, 10,000 images

Dashboard Design



Activity 1: Interactive and visually appealing dashboards

Potential Outcomes from the Dashboard

1. **Agricultural production is generally increasing**
The year-wise graphs show an overall upward trend in agricultural production from **1997 to 2021**, despite some fluctuations.
2. **Kharif crops contribute the largest share**
The seasonal distribution indicates that **Kharif production is much higher than Rabi and Autumn**, suggesting strong dependence on Kharif-season agriculture.
3. **Production is concentrated in a few states**
States such as **Uttar Pradesh, Tamil Nadu, Karnataka, and Andhra Pradesh** appear among the major contributors, indicating regional concentration of agricultural output.
4. **Rice occupies the largest cultivated area**
The crop-area comparison shows **Rice** having the largest cultivated area, followed by **Wheat and Cotton**, highlighting rice's major role in Indian agriculture.
5. **Changes in cultivated area may affect future production**
The state-wise cultivated-area comparison shows considerable differences between states. If cultivated areas decline in major producing regions, **future agricultural production could face pressure**, while better productivity could offset some of this effect.

Activity 2

Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

https://public.tableau.com/views/Book1_17874145556230/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Data Collection and Preprocessing Phase

Date	16August 2026
Team ID	----
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing: -

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	• Schema Inspection: Examined dataset attributes including, Season, Crop, Area (Hectares), and Production. • Dataset Profiling: Assessed missing values, structural irregularities, and value distributions across 30+ states, 600+ districts, and 50+ crop varieties from 1997 to 2021.
Data Cleaning	• Null Value Handling: Imputed missing values in Production using regional median yield rates; dropped records missing essential spatial identifiers. • Outlier & Noise Reduction: Removed corrupt

	entries with negative area values or impossible yield spikes caused by data entry errors
Data Transformation	Calculated the standardized productivity metric $\text{Yield (Tones/Hectare)} = \frac{\text{Production}}{\text{Area}}.$ Normalization: Grouped low-frequency minor crops into broad categories (e.g., Pulses, Oilseeds, Cereals) for macro-level trend analysis.
Data Type Conversion	• Numerical Casts: Explicitly converted Area and Production to floating-point numeric types to allow statistical aggregations. • Temporal & Categorical Casting: Formatted Crop Year into standard integer date format and converted categorical text fields into strings for dynamic visualization filtering.
Column Splitting and Merging	Standardization: Cleaned and merged inconsistent state/district naming variations resulting from geographical boundary redefinitions over time. Attribute Parsing: Split combined seasonal text attributes (e.g., combining climate tags) into standardized season labels (Kharif, Rabi, Summer, Whole Year).

Data Modeling	❏ Star Schema Architecture: Created a central Fact Tab linked to look up Dimension Tables. ❏ Relationship Building: Established primary-foreign key relationships (1:N) between dimensions and facts to optimize query speed and drill-down performance in Tableau.
Save Processed Data	❏ Storage Format: Exported clean data to optimized .csv / .parquet formats and pushed structured tables to SQL database staging tables. ❏ Tableau Extraction: Created a hyper-extracted data source file (.hyper) ready for rapid visualization render in

Data Collection and Preprocessing Phase

Date	13 August 2026
Team ID	
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	3 Marks

Data Quality Report:

Data Source	Data Quality Issue	Severity	Resolution Plan
Dataset 01	Numerical entries are absent.	Moderate	Filter dataset and impute absent numerical values.
Dataset02	Crop-wise separated production report	High	Make a file containing all reports

Date	12August2026
Team ID	-----
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	2 Marks

Data Collection Plan & Raw Data Sources Identification: -

Data collection:

Section	Description
Project Overview	The India's Agricultural Crop Production Analysis project constructs an interactive visual analytics platform to evaluate multi-decade crop output across 30+ states and 600+ districts (1997–2021). By converting raw metrics into Tableau dashboards, it provides actionable insights to help farmers, policymakers, and investors optimize yield, reduce climate risk, and direct resources effectively.
Data Collection Plan	Extract multi-year district and state agricultural records from official government open-data portals and academic repositories. Pull core spatial (State, District), temporal (Crop Year,

	Season), and output attributes (Area in Hectares) Ingest downloaded CSV/Excel data files into a local SQL database for record-count validation and initial schema normalization.
Raw Data Sources Identified	Raw Data Sources Identified • Ministry of Agriculture (DES / APY Corpus): Official Area, Production, and Yield statistics across 30+ states, 56+ crops, and 5 cropping seasons (1997–2021). •

Raw Data Source:

Source Name	Description	Location/URL	Format	Size	Access Permissions
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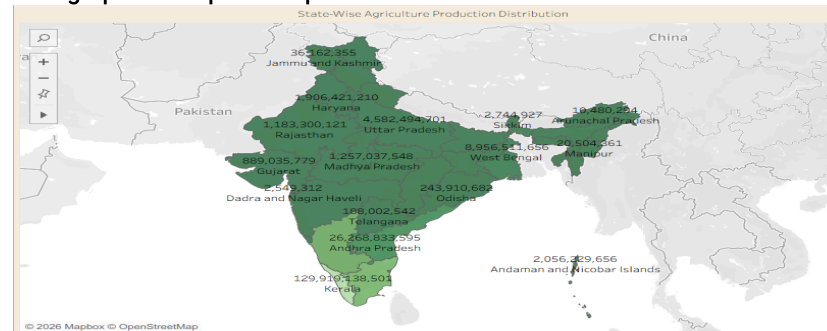
Dataset 1	Ministry of agriculture and farmers welfare.	https://agriwelfare.gov.in/	CSV format dataset	1GB	Public
Dataset 2	Smart bridge site	thesmartbridge.com	Excel	1 GB	Public
Dataset 3	Official statics compiled by the directorate of economics and statics	https://desagri.gov.in/	Excel	1GB	Public

Business Question and Visualization Report

Date	14 August 2026
Team ID	
Project Name	India's Agricultural Crop Production
Maximum Marks	5 Marks

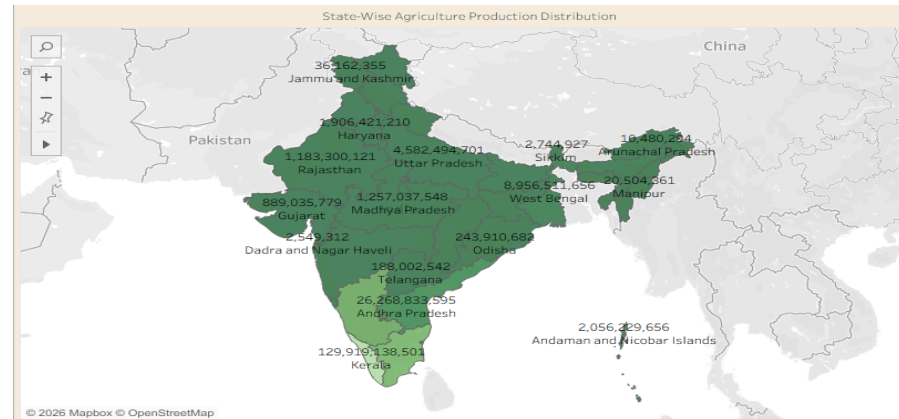
1. Which state has maximum production?

*Geographical map shows production of each state.

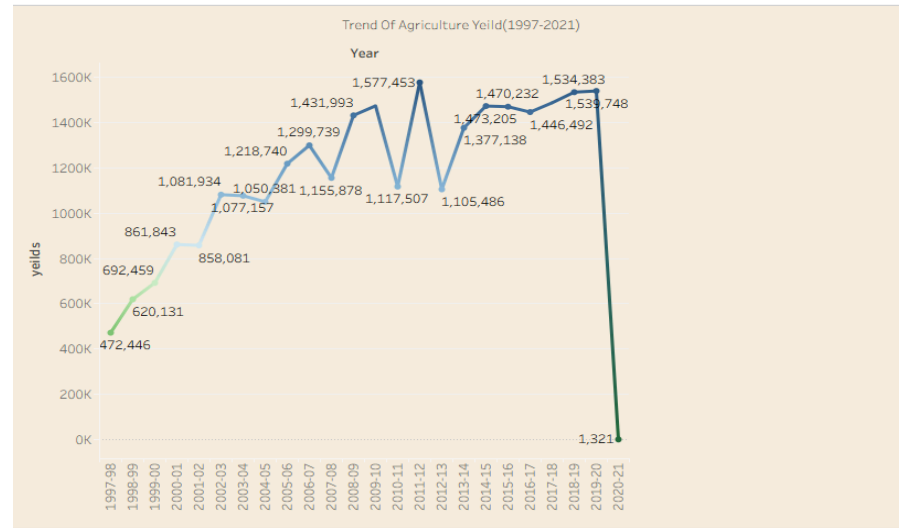


2. Which state has minimum production?

*Geographical map shows production of each state.



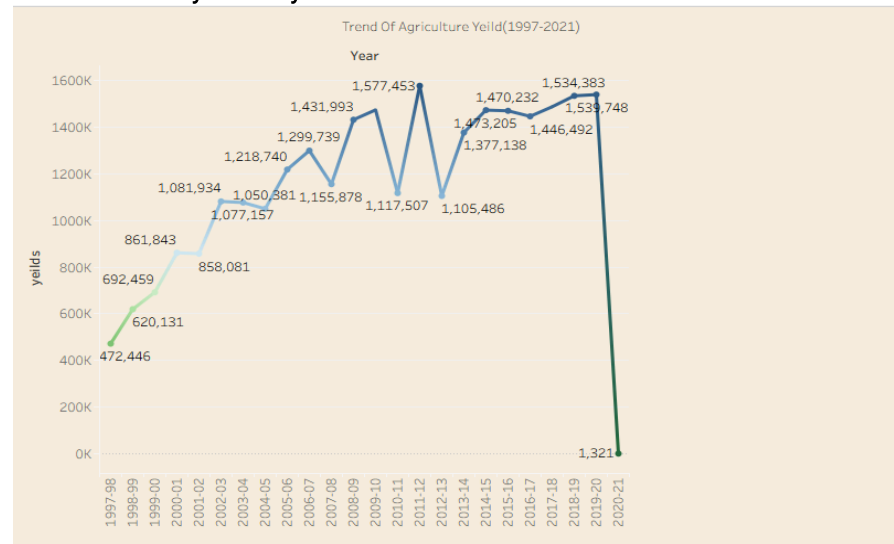
3. Which year has maximum yield?
 *bar chart shows year and yield



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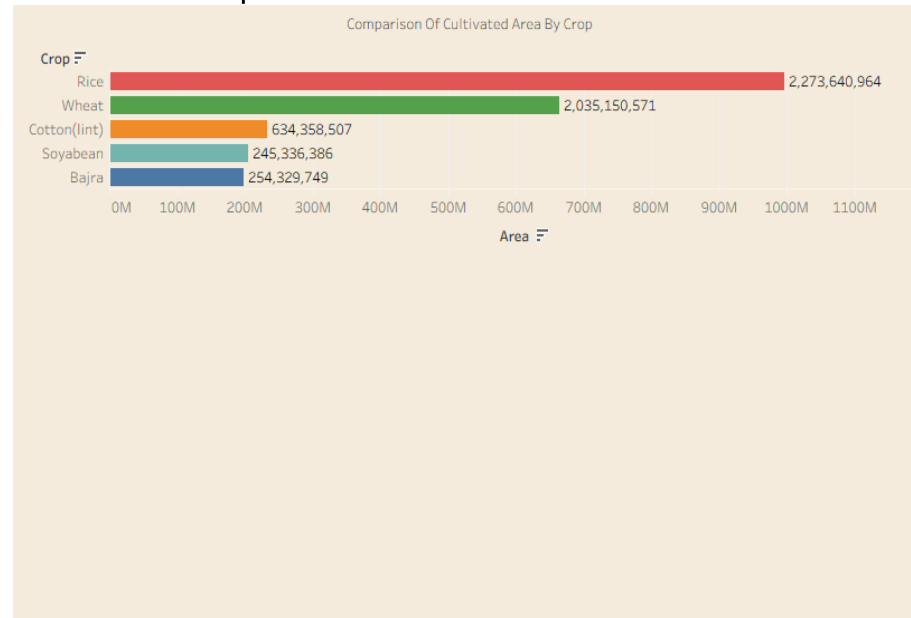
4. Which year has minimum yield?

*bar chart shows year and yield



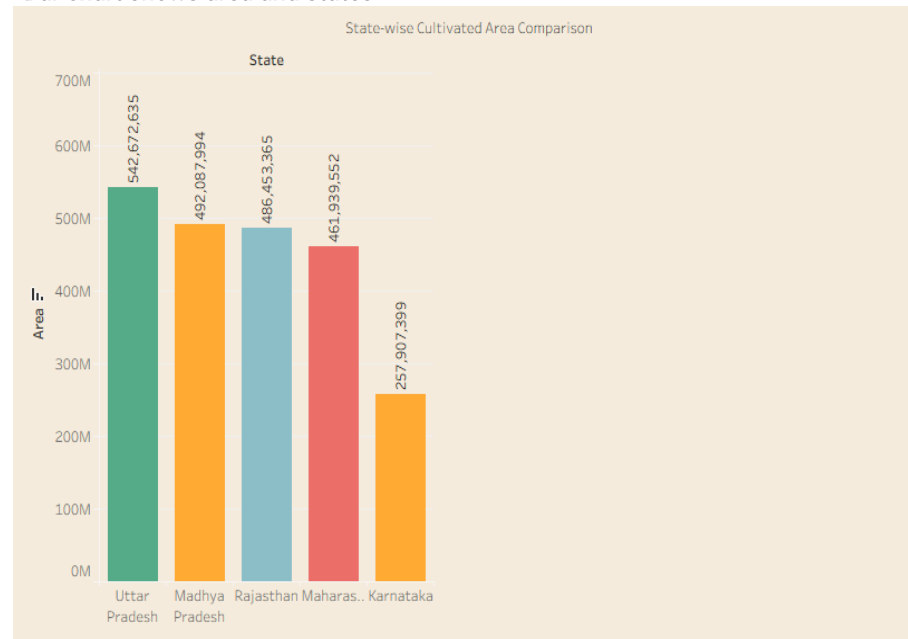
5. Which crop has maximum cultivated area?

*Bar chart shows crop and area



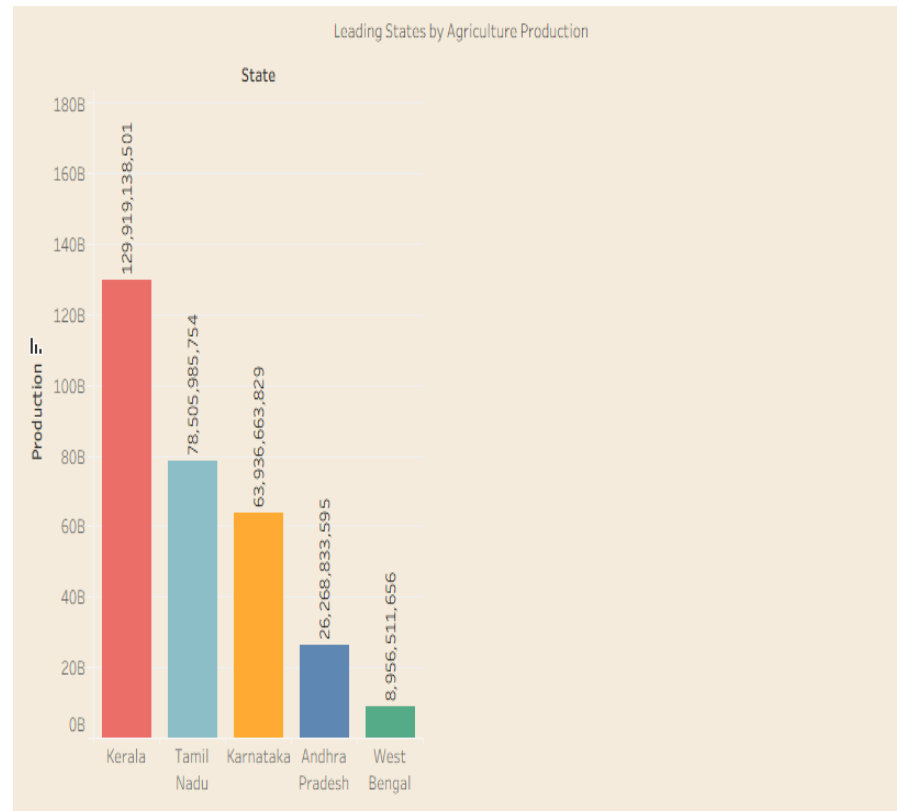
6. Which state has maximum cultivated area?

*Bar chart shows area and states



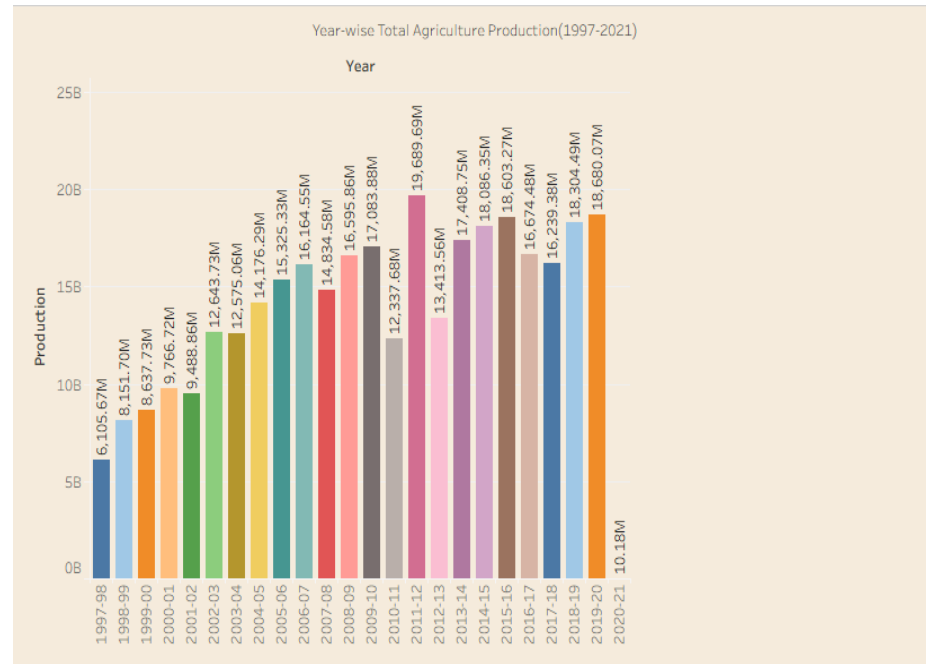
7. Which state has maximum agriculture production?

*bar chart shows leading states



8. Which year has maximum total agriculture production?

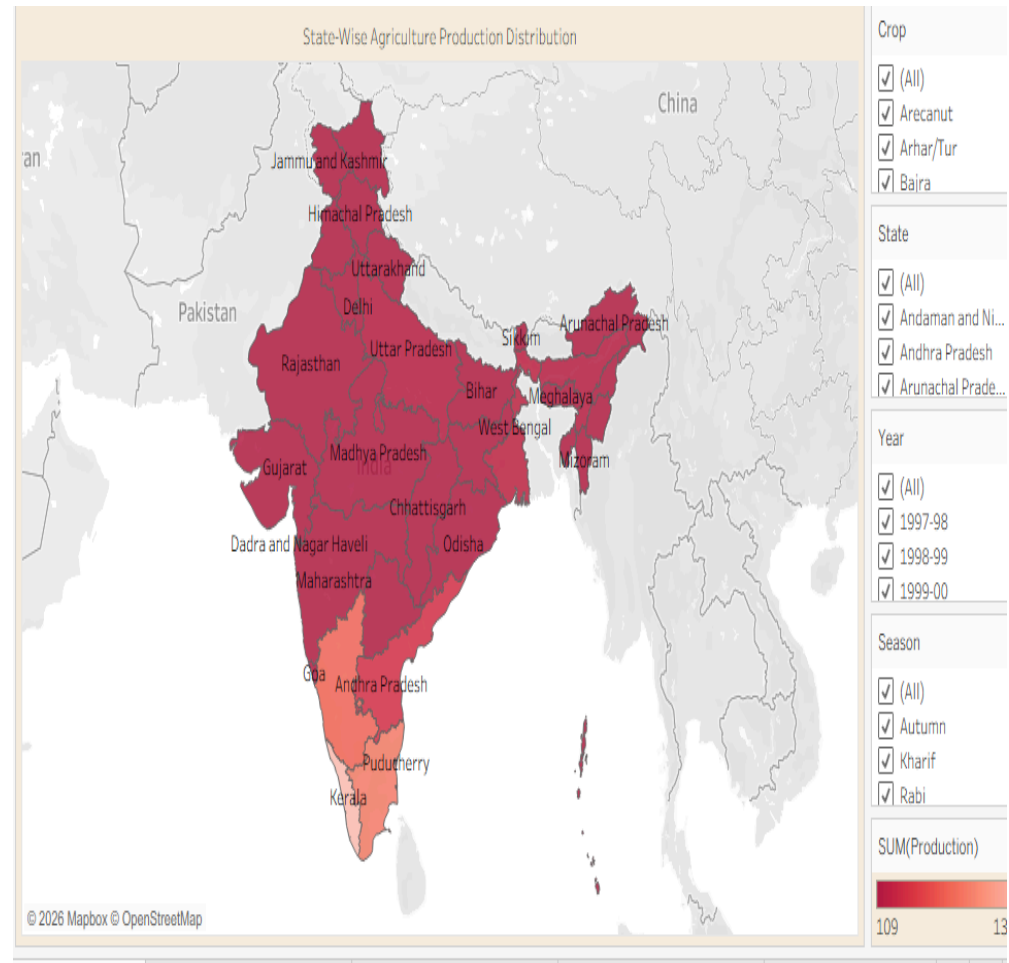
*bar chart shows year wise agriculture production.



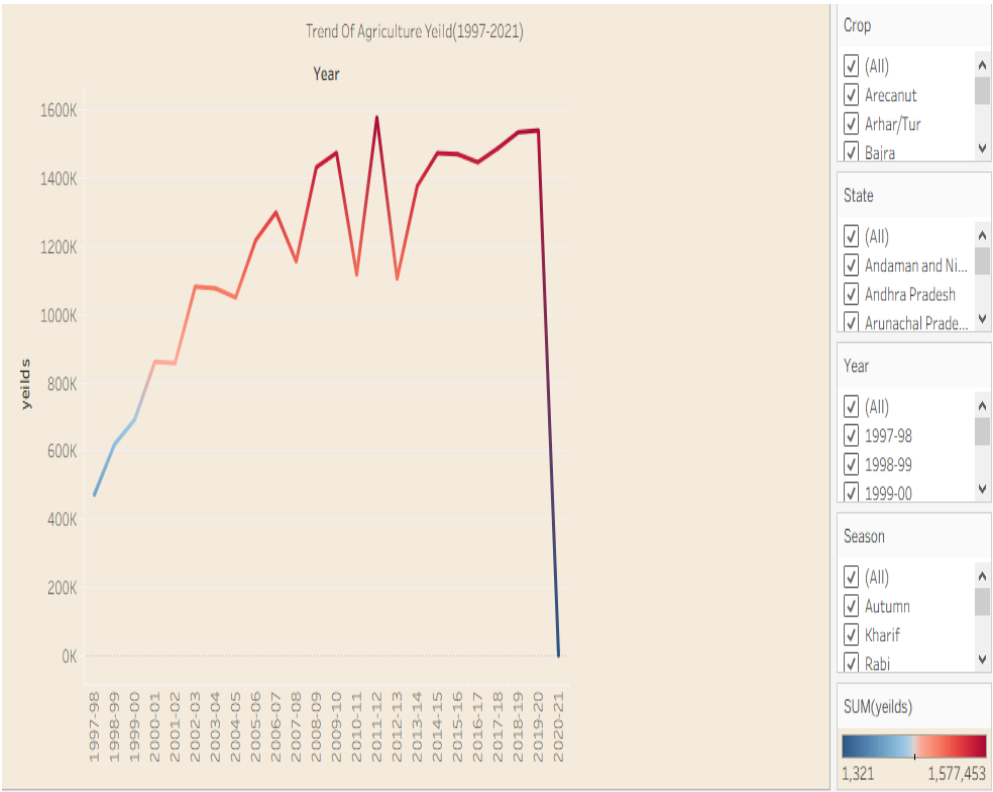
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4.2. Developing Visualizations

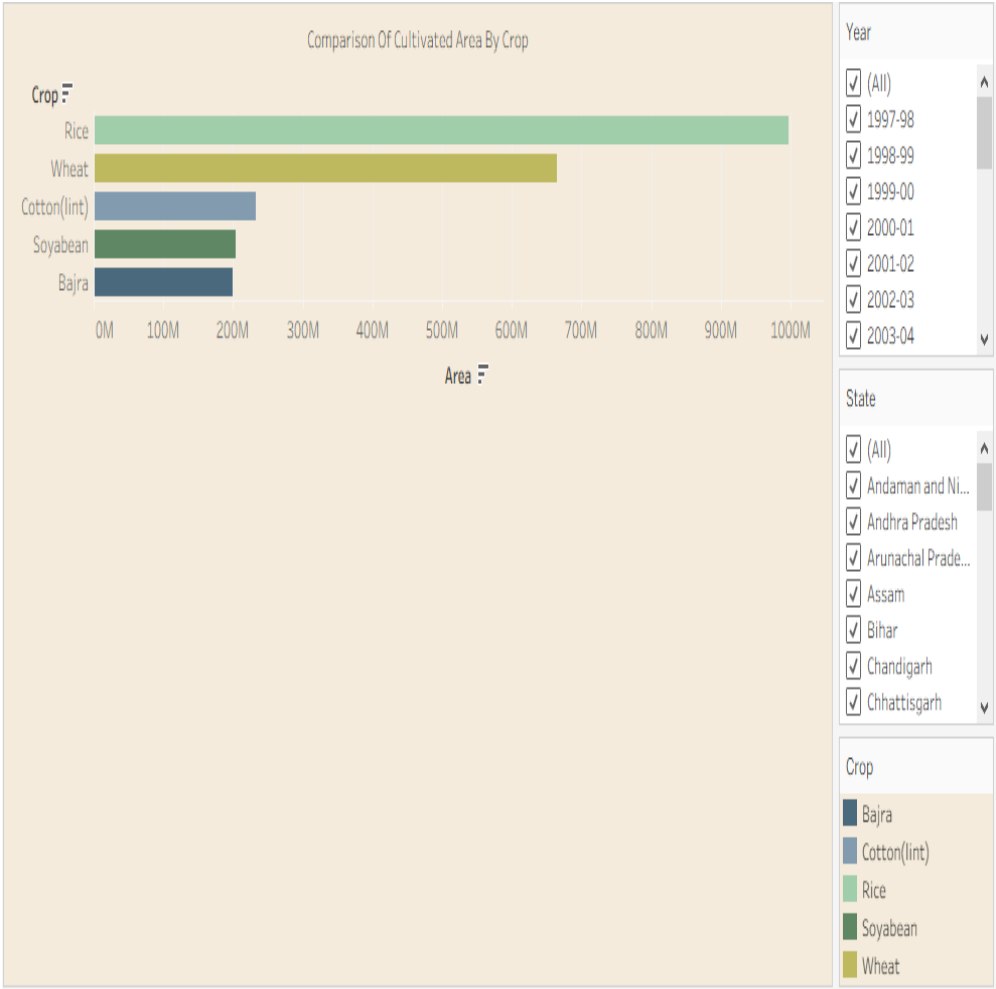
1.



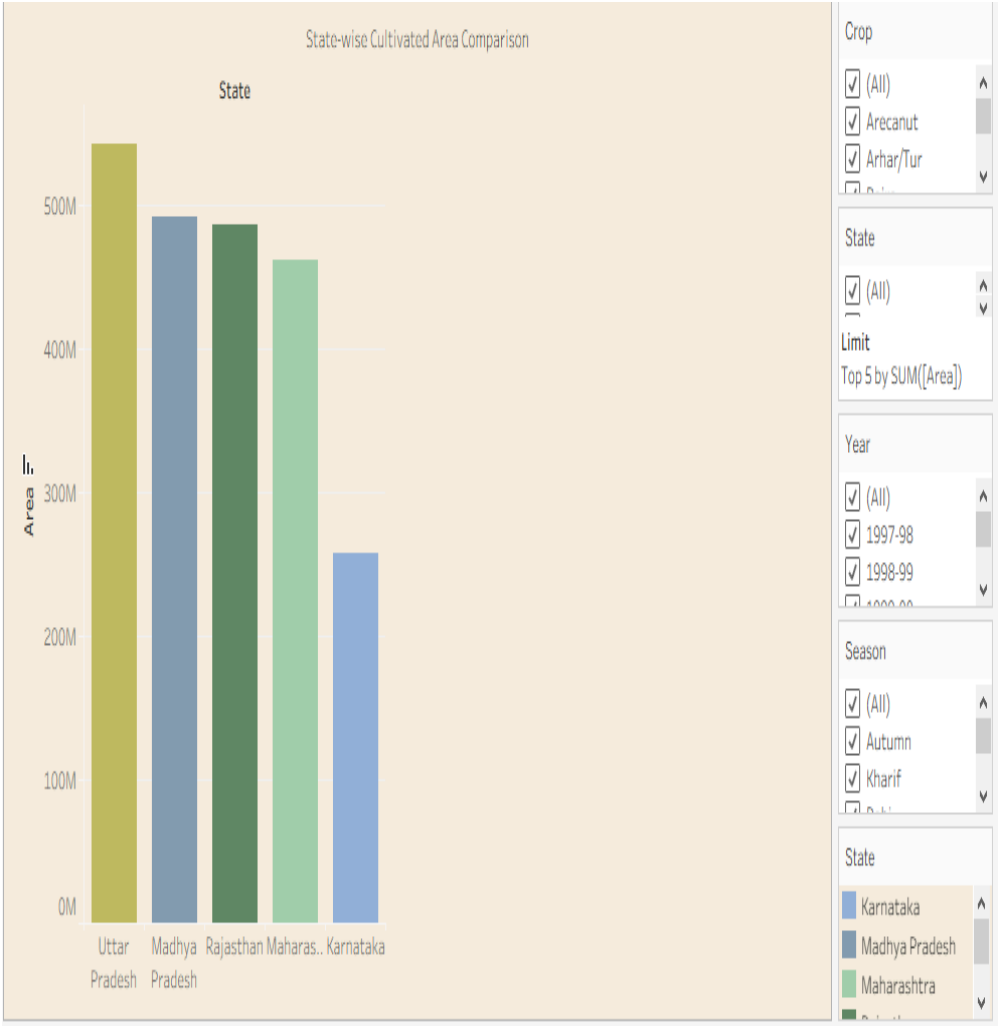
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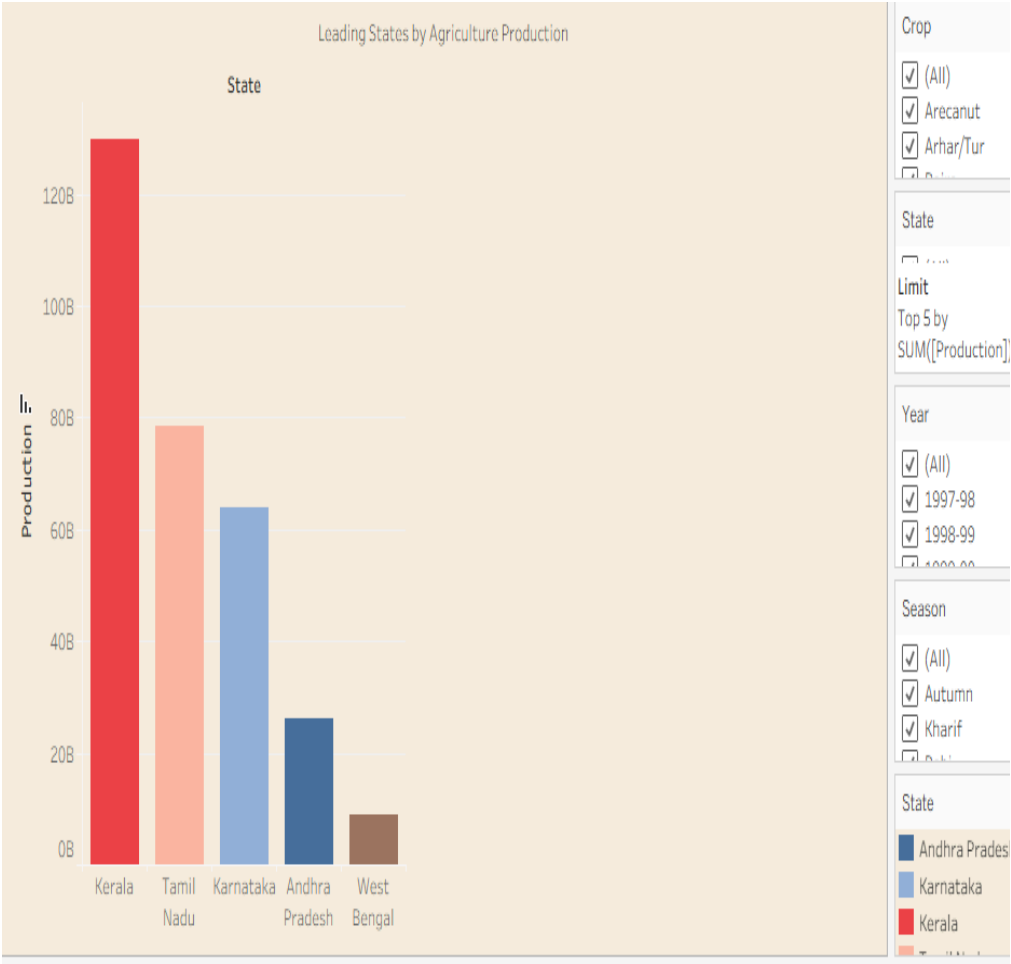
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4.



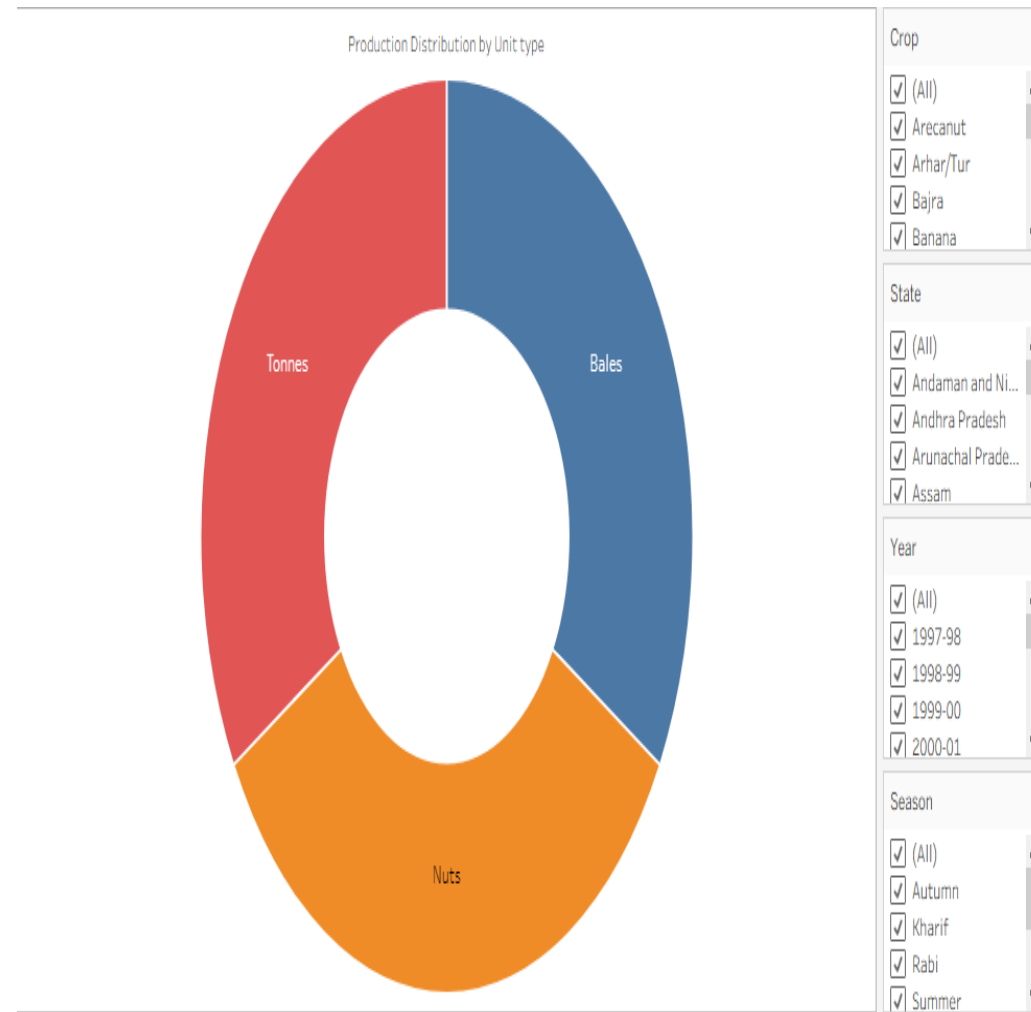
5.



6.



7.



8.

